



Wrap and Loft

Wrap curves onto the profile, offset them along its normal, and loft a strip -- sidewall and core-edge reference surfaces that follow rocker and camber.



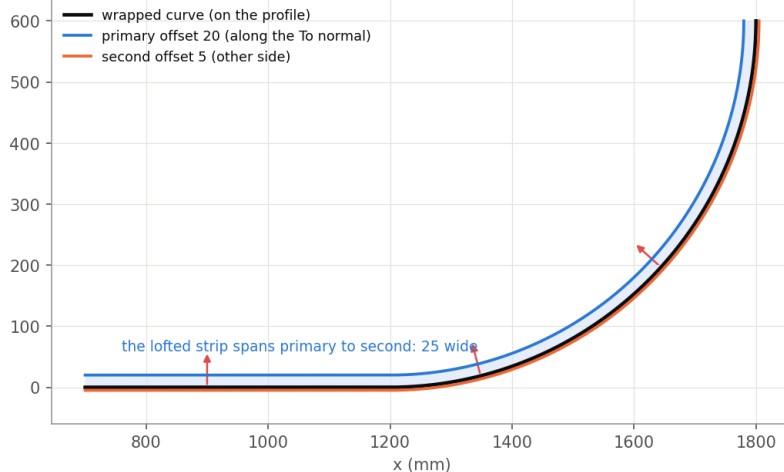
What it does

A reference surface that follows the profile through the thickness needs the wrapped curve and a matching offset curve, lofted -- with no gaps where source curves meet. Wrap and Loft does all of it in one feature.

Use it when

- Sidewall or core-edge reference surfaces
- A strip of set height along the profile
- Planar sources (the From path is made for you)

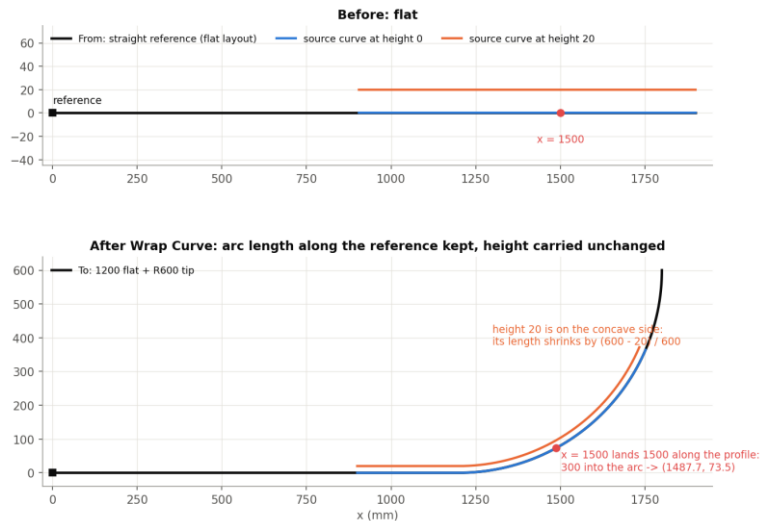
Wrap and Loft: offset the wrapped curve along the To normal, loft between the offsets



Primary 20 + second 5: a 25 wide strip along the To normal.



The bending map



Arc length kept

A point s along the From reference lands s along the To reference (a shift, no scaling).

Height and width kept

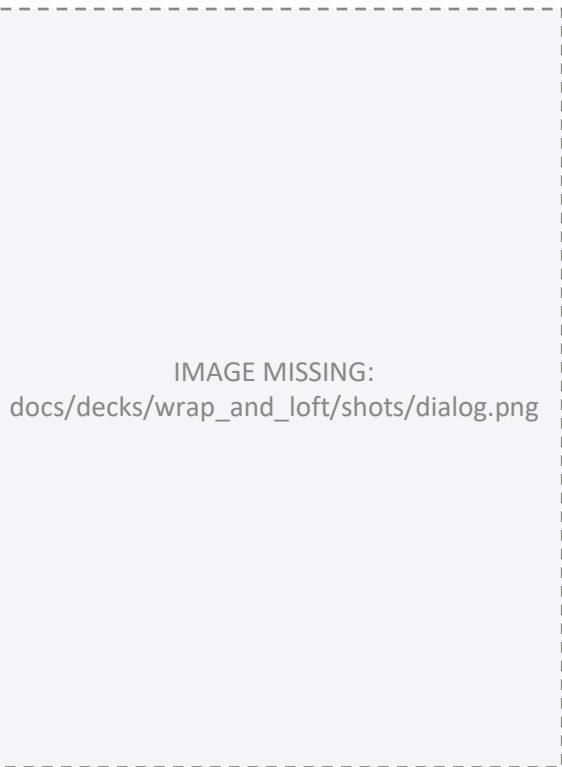
The point's offsets in the frame carry over: ribs stay perpendicular.

Layers bend

At height h around radius R they stretch or shrink by $(R - h) / R$, as a real layer does.



The dialog

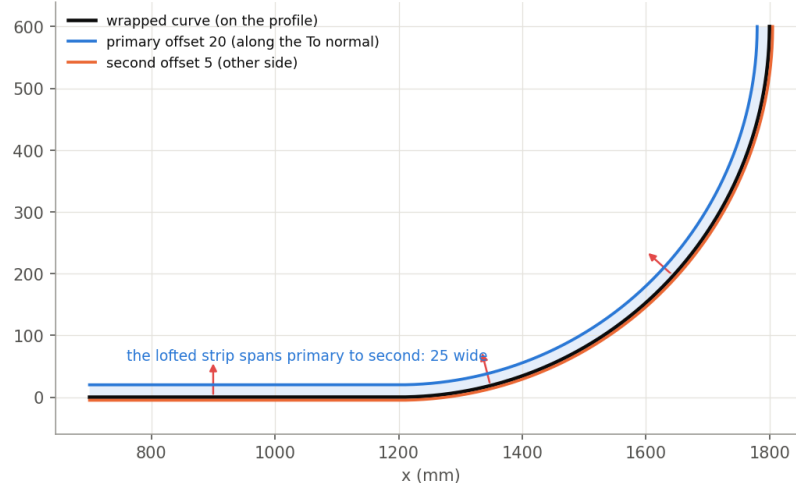


#	Parameter	What it means
1	Wrap edges	The source curves.
2	From data	From edge(s) (G1 chain) and From reference (vertex, mate connector, plane).
3	To data	To edge(s), Flip (reverse To), To reference, Flip normal.
4	Frame orientation	Frenet (default) / Binormal (face, mate connector or world axis).
5	Offset	Primary offset (20 mm), Flip offset direction, Second direction + Second offset.
6	Keep output curves	Keep all / Keep wrapped only.
7	Details	As Wrap Curve, + Fix control-point clustering (on), Break spans, Exact ruled surface.
8	Debug	Print curves, show frames.



What makes it robust

Wrap and Loft: offset the wrapped curve along the To normal, loft between the offsets



Planar shortcut

Planar sources: From edge(s) disappears; the From path is the To chain projected onto the source plane (wraps, not drapes).

Welds

Where curves meet, the later adopts the first's offset direction: no 0.35 mm gaps at seams.

Exact ruled (option)

An exact ruled B-spline instead of a loft where possible; changes face ids.



Outputs and messages

Outputs

Key	What it holds
(sheet)	One face per source curve, unioned.
(wires)	Optional: wrapped, primary, second curves.

Messages

	Message / what to do
Info	clustering fix recommended Only when it is off and slivers were found.



Tips and limits



Negative offset

Use Flip offset direction; primary cannot be negative.



Planar check

A single straight source edge cannot define a plane: the flag keeps its previous value.



Failures

Loft / union failures are reported on the console only.



Where it lives

Document	curveMapping_public (d12583277c42b00c63a9b7b7)
Tab	wrapAndLoft (+ engine curveMappingCore)
Tests	none
Explainer	docs/explainers/curve_mapping/curve_mapping_explained.md, section 2.2
Related	Wrap Curve, Wrap and Loft, Deform, Measure curve distance, Offset edges